UI/UX Documentation

AI-Powered Construction Site Intelligence Platform

Version 1.1 | Status: Design-Ready

1. Design Philosophy

This is a safety-critical operational dashboard, not a consumer product. Design decisions are governed by three principles:

Clarity over decoration. Every pixel must earn its place. Dense operational information is acceptable; decorative elements that add visual noise are not.

Alerts must be unmissable. Safety-critical alerts (PPE violations, zone breaches, escalated discrepancies) must be visually distinct through color, icon, and position — not just a subtle badge. A supervisor must be able to see an unacknowledged violation without scrolling or looking for it.

Honest system state. The UI must never allow "no alerts" to be ambiguous with "camera is down." System health is always visible so users know when to trust what they are seeing.

2. Information Architecture

The information hierarchy has three levels:

- Level 1 — Dashboard: aggregated site health at a glance. KPI cards, live alert feed, system status.
- Level 2 — Module lists: Violations, Deliveries, Materials, Equipment, Hazard Reports — tabular, filterable, sortable.
- Level 3 — Detail views: individual event detail with all metadata, actions (acknowledge, escalate, resolve), and evidence (screenshot, image).

Navigation is sidebar-based, persistent on desktop. Mobile collapses to a bottom navigation bar (read-only summary view only — full operational control is desktop-first).

3. Sitemap

Path	Screen Name	Roles
/login	Login	All
/dashboard	Main Dashboard	Admin, Supervisor
/cameras	Live Cameras	Admin, Supervisor
/violations	Violations List	Admin, Supervisor
/violations/:id	Violation Detail	Admin, Supervisor
/zones	Zone Management	Admin
/equipment	Equipment Tracker	Admin, Supervisor
/deliveries	Deliveries + Material Log	Admin, Supervisor
/deliveries/discrepancies	Discrepancy Escalations	Admin only
/deliveries/materials	Material Inflow Log	Admin, Supervisor
/hazards	Hazard Reports List	Admin, Supervisor
/hazards/new	Submit Hazard Report	Worker
/hazards/:id	Hazard Report Detail	Admin, Supervisor
/users	User Management	Admin only
/settings	Settings	Admin, Supervisor (limited)

4. Wireframe Descriptions

4.1 Login Screen

Centered card on a dark/gray background. Logo at top. Email field, password field, Login button. No self-registration link (admin-created accounts only). Error message appears inline below the form on failure — generic "Invalid credentials," no specifics.

4.2 Dashboard Screen

Top row: 4 KPI cards spanning full width — PPE Compliance Rate (%), Active Violations count, Today's Vehicle Count (AI-verified), Open Hazard Reports.

Second row: Live Alert Feed (left 60%) and System Status panel (right 40%) showing camera health per gate/zone.

Third row: Material Inflow Summary for today (admin/supervisor) — bar chart: material type vs. quantity received.

Escalation Queue banner (Admin only): amber/red banner across top of dashboard if unresolved discrepancies exist — unmissable.

4.3 Violations List Screen

Full-width table. Columns: Timestamp, Worker ID (if identified), Zone/Gate, Violation Type, Screenshot thumbnail, Status (Unacknowledged / Acknowledged). Row click → detail. Filter bar above: date range, violation type, zone, status.

4.4 Violation Detail Screen

Two-column layout. Left: violation screenshot with bounding-box overlay (worker outline + PPE status labels). Right: metadata panel — timestamp, gate/zone, violation type, worker ID, detected confidence score, current status, Acknowledge button.

4.5 Log Delivery Form

Simple form with clearly labeled fields: Gate (dropdown), Manual Vehicle Count (number input, min 0), Material Type (dropdown: Cement / Sand / Bricks / Steel / Pipes / Other), Quantity (decimal input), Unit (dropdown: Bags / Tons / Cubic Meters / Pieces). All fields required. Submit button. Note below form: "AI vehicle count is recorded automatically at the gate — your entry is matched against it."

4.6 Discrepancy Detail Screen (Admin only)

Side-by-side comparison card: AI Vehicle Count (left, blue) vs. Manual Reported Count (right, red if mismatch). Delta highlighted. Below: details — Gate, Date, Time, Supervisor who filed (name + ID), Material Type and Quantity logged (informational). Action buttons: Resolve / Flag for Investigation / Request Explanation. Timeline at bottom: history of all deliveries at this gate in the past 30 days, mismatches highlighted.

4.7 Material Inflow Log Screen

Date-range filter + material type filter at top. Table: Date, Time, Gate, Material Type, Quantity, Unit, Logged By (supervisor name). Sortable by any column. Export to CSV button (admin). Summary row at bottom: total per material type for selected period.

4.8 Hazard Report Submission (Worker)

Full-screen form (workers see nothing else). Image upload (drag and drop or tap to select). Description text area. Location field (text — e.g., "North scaffolding section, Level 3"). Submit button. On success: "Your report has been submitted. Category: [AI-assigned category]." Workers cannot see other reports.

5. Screen List Summary

Screen	Primary Action	Key Components
Login	Authenticate	Email + password form, error inline

Screen	Primary Action	Key Components
Dashboard	Monitor site	KPI cards, alert feed, system status, material summary
Violations List	Review violations	Filterable table, status badges, thumbnail
Violation Detail	Acknowledge	Screenshot with overlay, metadata, action button
Zone Management	Define zones	Map/polygon editor, zone type dropdown
Equipment Tracker	Monitor machinery	Table: type, location, idle time, last seen
Deliveries	Log deliveries	Log form, delivery history table
Discrepancy Escalations	Resolve fraud flags	Comparison card, history timeline, action buttons
Material Inflow Log	Review materials	Filterable table, totals row, CSV export
Hazard Reports List	Manage hazards	Table with AI category badges, status
Hazard Submission	Report hazard	Image upload, description, location, submit
User Management	Manage users	Table, add/edit modal, deactivate button

6. UI Components

6.1 KPI Card

White card, 8px border radius, subtle drop shadow. Icon (top-left, colored per status). Metric value (large, bold, 32px). Label (small, gray, 12px). Optional trend arrow. Color-coded left border: green (good), amber (warning), red (critical).

6.2 Status Badge

Pill-shaped inline label. Variants: Compliant (green), Violation (red), Pending (amber), Acknowledged (gray), Escalated (orange), Resolved (green). Used in tables and detail screens.

6.3 Alert Toast

Slides in from top-right. Icon + message + timestamp. Color: red for violations/escalations, amber for warnings, blue for informational. Requires explicit dismiss click — does not auto-disappear for safety-critical alerts.

6.4 Data Table

Sticky header. Zebra striping (white / gray-50). Column sorting via header click. Row click opens detail. Pagination controls at bottom (20 rows default). Empty state: "No records found" with filter-reset suggestion.

6.5 Modal

Centered overlay with backdrop. Used for: Add User, Confirm Action (resolve/flag), Zone definition. Always has explicit Close and primary action buttons.

7. Design System

7.1 Typography

Use Case	Font	Size	Weight
Page title	Inter	28px	Semi-bold (600)
Section heading	Inter	20px	Semi-bold (600)
Card metric value	Inter	32px	Bold (700)
Body text	Inter	14px	Regular (400)
Table row	Inter	13px	Regular (400)
Label / caption	Inter	12px	Medium (500)
Monospace (IDs, timestamps)	JetBrains Mono	13px	Regular (400)

7.2 Color Palette

Purpose	Color Name	Hex
Primary action	Blue-600	#2563EB
Primary hover	Blue-700	#1D4ED8
Success / Compliant	Green-600	#16A34A
Warning / Pending	Amber-600	#D97706
Danger / Violation	Red-600	#DC2626
Escalation highlight	Orange-500	#F97316
Page background	Gray-50	#F9FAFB
Card background	White	#FFFFFF
Sidebar background	Gray-900	#111827
Sidebar text	Gray-100	#F3F4F6
Body text primary	Gray-900	#111827
Body text secondary	Gray-500	#6B7280
Border / divider	Gray-200	#E5E7EB

7.3 Icons

Icon set: Lucide React (open source, consistent stroke weight). Key icons:

Icon	Usage
hard-hat	PPE / helmet compliance
alert-triangle	Violation / warning
truck	Delivery / vehicle count
package	Material inflow
map-pin	Zone / restricted area
camera	Live camera feed
shield-check	Compliant status
shield-x	Non-compliant / violation
wrench	Equipment
flag	Hazard report
user	Worker / user
bar-chart	Analytics / KPI
log-out	Logout

7.4 Buttons

Variant	Use Case	Style
Primary	Main action (Submit, Log Delivery, Acknowledge)	Filled Blue-600, white text, 8px radius, 40px height
Secondary	Alternative action (Cancel, Back)	Outlined Blue-600, transparent background
Danger	Destructive (Flag for Investigation, Deactivate User)	Filled Red-600, white text
Ghost	Tertiary (Export, Filter)	No border, Gray-700 text, hover background
Disabled	Unavailable action	Gray-300, not clickable, no hover

7.5 Forms

- Label sits above input field — never placeholder-only (accessibility).
- Required fields marked with asterisk (*) in label.
- Inline validation error appears in Red-600 below the field on blur.
- Dropdowns used wherever input is constrained (material type, unit, gate, zone type) — prevents dirty data.
- Submit button disabled until all required fields are valid.
- File upload: drag-and-drop zone with dashed border + "Click to upload or drag here" label. Shows filename + size after selection. Accepted types: JPG, PNG, max 10MB.

7.6 Tables

- Sticky header on scroll.
- Zebra striping: white / Gray-50 alternating rows.
- Status badges inline in relevant column.
- Screenshot thumbnails (48x48px) in violations table — click to expand.
- Row hover: Gray-100 background.
- Row click: navigate to detail view.
- Empty state: centered message + icon + reset filter link.

8. Responsive Design

Breakpoint	Behaviour
Desktop (≥1280px)	Full sidebar + main content. All tables, charts, and detail panels visible.
Tablet (768–1279px)	Sidebar collapses to icon-only. Tables scroll horizontally. Detail panels stack vertically.
Mobile (<768px)	Bottom navigation bar. Read-only summary cards only. No delivery logging, no zone management, no user management on mobile.

Note: The platform is designed desktop-first. Mobile is read-only intentionally — operational actions (logging deliveries, managing zones, acknowledging violations) require a keyboard and full-screen view for accuracy.

9. Accessibility Guidelines

- WCAG 2.1 AA compliance minimum.
- Color contrast ratio: minimum 4.5:1 for body text, 3:1 for large text and UI components.
- All icons paired with visible text labels — no icon-only buttons for critical actions.
- All form inputs have associated <label> elements — not just placeholder text.
- Alert toasts include role="alert" for screen-reader announcement.
- Tables include proper <th> scope attributes and caption elements.
- Keyboard navigation: all interactive elements reachable and usable via Tab/Enter/Space.
- Focus indicator: visible blue outline (2px, Blue-600) on all focused elements — never remove outline without replacement.

10. UX Best Practices

- Critical alerts (violations, escalations) always visible above the fold on dashboard load — never require scrolling to see an unacknowledged safety alert.
- Never auto-dismiss safety alerts — all violation and escalation alerts require explicit acknowledgment click.
- System/camera health status shown persistently in sidebar footer — "no alerts" is never ambiguous with "camera is offline."
- Delivery log form makes clear that AI count is independent — "AI vehicle count is recorded automatically. Your manual entry is compared against it." — so supervisors understand the system's fraud-detection mechanism.
- Discrepancy escalation screen shows the material type logged (informational), but makes clear the discrepancy flag is based on vehicle count only — not material — to avoid confusion.
- Empty state screens always explain why they are empty and suggest an action — never a blank screen.
- Destructive actions (deactivate user, flag for investigation) require a confirmation modal — never single-click.

11. Micro-interactions

Trigger	Animation	Purpose
New violation alert arrives	Toast slides in from top-right (300ms ease-out)	Draw attention to new critical event
Live camera active	Pulsing green dot (1.5s loop)	Confirm feed is live, not frozen
KPI card on dashboard load	Count-up animation from 0 to actual value (600ms)	Make metrics feel dynamic, not static
Gate status: offline	Red pulsing border on camera card	Visually distinct from "no events" vs "no feed"
Form submit in progress	Button shows spinner + "Saving..." text, disabled	Prevent double-submission
Row acknowledged	Status badge transitions from red to gray (200ms fade)	Confirm action completed without page reload
Sidebar active item	Left border highlight + background fill on active link	Clear current location indicator
Discrepancy banner dismiss	Slide up and fade (if resolved), not just disappear	Action feels confirmed, not glitchy